

(15 pts) Use the matrix A and its reduced row echelon form to answer each part. In each case give a reason for your answer.

$$A = \begin{bmatrix} 1 & -2 & 4 & 2 \\ -2 & 4 & -8 & 2 \\ 0 & 1 & -3 & 3 \end{bmatrix} \quad \text{RREF}(A) = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a) Are the columns of A linearly independent?

No. There is not a pivot in every column.

b) Do the columns of A span $\mathbb{R}^3$?

Yes. There is a pivot in every row.

c) Give a geometric description of the solution set for the homogeneous system $A\mathbf{x} = \mathbf{0}$.

Since there is one free variable, the solution to the homogeneous system is a line through the origin in $\mathbb{R}^4$ in the direction of the vector $\begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}$.

d) (1 pt) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the matrix transformation defined by $T(\mathbf{x}) = A\mathbf{x}$. What are the values of m and n ?

$$m = 3 \quad n = 4$$

e) Is the vector $(1, -2, 4, 2)$ in the range of T ? No. vectors in the range of T must be in $\mathbb{R}^3$ not $\mathbb{R}^4$.

f) Write the product $A \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$ as a linear combination of the columns of A .

$$1 \begin{bmatrix} 1 \\ -2 \\ 4 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} -2 \\ 4 \\ -8 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} 4 \\ -8 \\ -3 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

g) Is the transformation T one-to-one?

No. There is a free variable.

h) If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, list any free variables in the system $A\mathbf{x} = \mathbf{b}$.

x_3 is the only free variable